Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Screening of Heterogeneous Second-Life EV Batteries

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***Abstract*—Retired electric-vehicle batteries can be reused in less demanding applications, but a health model trained on one battery population may fail on the next. Continual learning promises to adapt to each new population without forgetting earlier ones. We test this promise for forecasting the state of health (SOH) of lithium-ion cells one to six capacity checks ahead, on a leakage-controlled benchmark from five public datasets (395 cells), with pre-registered follow-up experiments and every claimed difference replicated on a second seed set. A Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) is compared with fine-tuning, replay, joint training, literature backbones and simple forecasters. Forgetting appears whenever usage conditions differ (up to 0.46 percentage points, pp). The proposed Residual TFT-EWC learns a shrunk correction to linear extrapolation and holds up where each traditional forecaster fails: on the continual stream it matches extrapolation and beats gradient boosting by 0.13–0.41 pp; on unseen usage conditions it beats extrapolation by 0.43–0.68 pp, but gradient boosting with all conditions stored, and the same TFT trained jointly, are more accurate. At this data scale, learning continually costs more than it gains. The proposed model's forecasts give the measured capacity-based health class for 95.5% of unseen-condition samples (gradient boosting 97.7%). Shift-adaptive EWC, first-life pretraining and Spark-extracted ageing-cycle inputs do not help. Forecast intervals reach nominal coverage on a new condition only when calibrated on conditions held out from fitting, and not outside the training range.**

***Index Terms*—Continual learning, second-life batteries, electric vehicles, state of health, battery degradation, heterogeneous batteries, catastrophic forgetting, machine learning**

I. Introduction

An electric-vehicle (EV) battery that no longer meets automotive requirements can often still serve as stationary storage or backup power. Reuse extends its life and delays recycling, but only if its health is judged reliably.

The state of health (SOH) describes a battery's present capability relative to a reference condition. Machine-learning models have estimated it from voltage, current, temperature, partial-charging and impedance measurements [1]–[4]. Retired batteries, however, differ in chemistry, manufacturer, capacity, first-life history, temperature and charging protocol, and their first-life records are often incomplete, so a model trained once may fail on the next population. Continual learning updates a model on new data while preserving earlier knowledge, and must be tested on old and new groups alike to reveal forgetting.

This work combines continual learning with health forecasting and a first screening step for reuse. We *forecast* the SOH one to six capacity checks ahead from everything measured up to the latest check: on the first dataset, estimating the current SOH from cycling features was beaten by simply carrying the last measured SOH forward (§III), and a reuse decision depends on future, not only present, health. The forecast is mapped to capacity-based classes that support, but do not replace, a full reuse assessment.

Our contributions are as follows (Fig. 1):

- A leakage-controlled benchmark for forecasting the SOH of second-life cells from five public datasets (395 cells after quality rules), with two label artefacts removed: test currents set from the previous capacity, and pulse charge counted in the capacity.
- An evaluation of a Temporal Fusion Transformer with Elastic Weight Consolidation (TFT + EWC) against fine-tuning, replay, joint training, simple forecasters and the model families used in the literature (BiLSTM, TCN, Transformer, GPR, random forest, SVR) under three protocols (a continual stream, unseen groups, first-life pretraining), reporting adaptation, forgetting and retention.
- Residual TFT-EWC, a continual learner that corrects linear extrapolation rather than replacing it. On the stream it matches extrapolation, which no model beats, and outperforms gradient boosting; on unseen usage conditions it outperforms extrapolation but not gradient boosting or its jointly trained form (replicated).
- Forecast intervals for unseen usage conditions: calibrated on the training conditions they cover too little, whereas calibrating on conditions held out from fitting restores nominal coverage, except for a condition outside the training range.

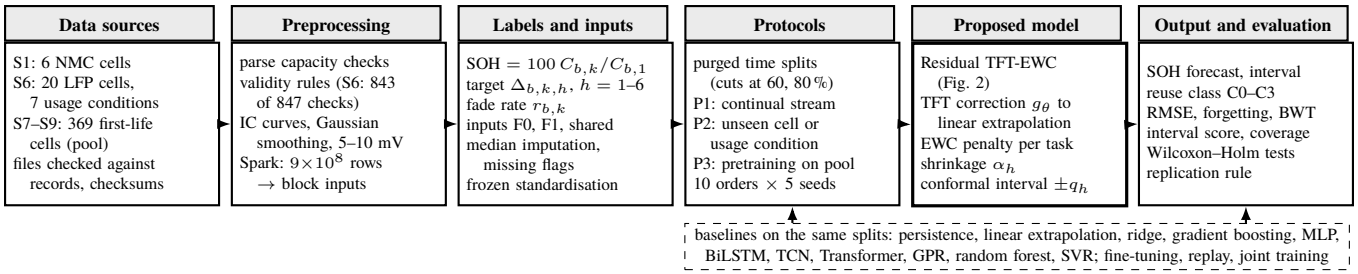


Fig. 1. System architecture: from the public archives to the forecast, its interval and the reuse class, with the protocols and baselines used for evaluation.

- A reproducibility analysis: several significance results on small battery datasets depend on the numerical environment, so the main results must pass a replication rule.

II. Related Work

We review studies from 2022–2026 (Table I; review in [5]). Second-life studies estimate the *current* SOH of a fixed population from partial charging under residential and fast-charging profiles (RMSE about 0.6 and 0.8%, with 85% less testing time than a capacity test) [1], from 58 health indicators of reused Nissan Leaf modules (about 0.2% RMSE, though the most useful indicators changed with ageing) [2], from impedance spectra [3], with bagged Gaussian process regression for screening (worst-case error below 1.48%) [4], and online with clustering for Leaf cells aged over 15 months (error below 2.3%) [6]. None adapts a model as new populations arrive.

Continual learning has predicted capacity degradation [7] and, with progressive networks and TCNs, the state of charge across drive cycles, addressing forgetting and task order [8]. Closest to our setting, pulse-test conditions have been learned as domain-incremental tasks with a replay buffer to estimate second-life SOH with near-zero forgetting [9]. Domain generalisation [10] and probabilistic BiLSTM/TCN models with isotonic recalibration [11] target second-life SOH without a task sequence.

Research gap. Continual learning for second-life SOH has so far targeted *estimation* of the current SOH [9]. To our knowledge, it has not been evaluated for *forecasting* across heterogeneous second-life populations with reuse screening, measured forgetting and comparison with simple forecasters and joint training. We ask whether a continual learner can learn such populations in sequence while retaining earlier ones, and whether its forecasts support reuse screening.

TABLE I

Approaches of the reviewed studies and of this work. Their errors are not comparable with ours (other cells, targets and SOH definitions).

Study	Approach	Predicts	CL	Reuse
[1]	partial-charge features	current SOH, second life	–	–
[2]	58 health indicators	current SOH, second life	–	–
[3]	impedance (EIS) + ML	current SOH, second life	–	–
[4]	statistical features, bagged GPR	current SOH, screening	–	✓
[6]	online ML, clustering	current health, second life	–	–
[7]	bilevel-optimised CL	capacity degradation	✓	–
[8]	PNN + TCN	state of charge	✓	–
[9]	pulse features, replay CL	current SOH, second life	✓	–
[10]	domain generalisation	current SOH, second life	–	–
[11]	BiLSTM, TCN, TCN-GPR	SOH with uncertainty	–	✓
This work	residual TFT + EWC, shrinkage, conformal intervals	SOH 1–6 checks ahead, interval	✓	✓

III. Methodology

We forecast the SOH one to three checks ahead (up to six in follow-up experiments) from everything measured up to a reference performance test (RPT, or capacity check), with cells or usage conditions arriving as a stream of tasks. The study first set out to estimate the *current* SOH, but on the first dataset cycle-level features carried almost no health information (median within-cell Spearman correlation with SOH 0.09, against -0.99 for elapsed time), and learned estimators were less accurate than persistence (1.78 and 2.37 pp for gradient boosting and ridge, against 0.99 pp). We therefore reframed the task after seeing the data, fixing the forecasting protocol before fitting any forecasting model.

A. Datasets

Table II lists the five public datasets (all CC BY 4.0). Every file is verified against its repository record or checksums, and our parsing of the raw files reproduces exactly the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks.

S1 contains six LG Chem INR21700-M50T cells that spent 23 months under an EV driving profile [22] and then about 27 months on peak-shaving duty cycles [23]; each RPT is a C/20 discharge interrupted by C/2 pulses, with impedance spectroscopy (EIS) from the fifth RPT [12]. Data-quality findings are handled by fixed rules (e.g. RPT 2, run at about 20 °C, is dropped).

S6 was selected by criteria fixed before screening (licence, a capacity-check series, ageing past first-life end of life, heterogeneity that S1 lacks). Its cells were checked every 100 cycles and end at 32–64% SOH; "second life" means continued ageing below 80% SOH under the same protocol. A check is used only if its C/25 discharge starts from full charge and reaches the cut-off (843 of 847). From the first-life

TABLE II
DATASETS. S1 AND S6 FORM THE CONTINUAL-LEARNING STREAMS; S7–S9 FORM THE FIRST-LIFE PRETRAINING POOL.

Set	Cells and conditions	Label	Samples
S1 [12], [13]	6 NMC 21700 cells, second life on residential or commercial peak shaving, 20/35 °C blocks	C/20 RPT capacity, pulses excluded	279
S6 [14], [15]	20 LFP 18650 cells, 7 conditions (charge mode, voltage and rate, discharge, depth), 50 °C	C/25 check capacity	2,409
S7 [16], [17]	195 of 228 NMC/C+SiO cells, calendar, cyclic and WLTP ageing, 0–40 °C	check-up capacity	10,251
S8 [18], [19]	55 NCM cells, six charge/discharge protocols	capacity cycles	9,699
S9 [20], [21]	119 of 130 NCA, NCM and blended cells, 25–45 °C	capacity cycles	16,761

sets *S7–S9*, cells with at least five valid checks are kept (369 cells, 36,711 samples).

B. Big-data processing of the ageing cycles

Between checks, every S6 cell logged each ageing cycle at about one sample per second (827 files, 132 GB, 9.0×10^8 rows). Apache Spark 4.2 (PySpark, local mode) reduces them to the mean, last value and slope of seven measured responses per 100-cycle block (21 inputs), joined to the check that ends the block; the job and its validation against an independent pandas implementation are in the supplement.

C. State of health and forecasting target

For cell b with check capacities $C_{b,1}, C_{b,2}, \ldots$ in time order,

$$\mathrm{SOH}_{b,k} = 100\, C_{b,k}/C_{b,1}\ \%, \tag{1}$$

relative to the first valid measurement, not a nominal rating. In S1 the capacity is the charge of the C/20 steps only [12, Eq. (1)]; the processed tables also counted the pulse charge, shifting some labels by up to 0.09 pp.

A sample is a triple (cell b, origin k, horizon $h \in \{1, 2, 3\}$), about one to three months in S1 and 100–300 cycles in S6. The target is

$$\Delta_{b,k,h} = \mathrm{SOH}_{b,k+h} - \mathrm{SOH}_{b,k}, \tag{2}$$

so an error in Δ equals the error in the future SOH; errors are root-mean-square errors (RMSE) in SOH percentage points (pp). The fade rate $r_{b,k}$ is the least-squares slope of SOH against time t (days or cycles) over the last three valid checks,

$$r_{b,k} = \frac{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})(\mathrm{SOH}_{b,i} - \overline{\mathrm{SOH}})}{\sum_{i=k-2}^{k}(t_{b,i} - \bar{t})^2}. \tag{3}$$

Inputs use only information available at the origin. **F0** contains the SOH history ($\mathrm{SOH}_{b,k}$, time or cycles since the first check, $r_{b,k}$), static descriptors (S1: duty cycle and first-life history [12]; S6: test conditions) and the *planned* usage until the target (S1: days at each temperature set point and at rest; S6: cycles), so forecasts are conditional on the planned usage. **F1** adds diagnostics measured at the origin: 47 in S1 (pulse resistances, incremental-capacity and EIS features) and 17 in S6 (plateau charges, incremental-capacity peaks, hysteresis, a resistance proxy). No test current is an input: in S1 the RPT and monthly currents were set from the previous capacity and so encode the label. Missing values are imputed from training data and flagged.

D. Splits and protocols

Purged chronological splits. Each cell's checks are cut at 60% (A) and 80% (B). Training targets lie at or before A; validation samples start at or after A and end by B; test samples start at or after B. Samples straddling a cut are dropped, so no training label falls in the validation period and no validation label in the test period.

P1, continual stream. A task is a cell in S1 (six tasks) or a test condition in S6 (seven tasks). At step i of an ordering π, a strategy receives only task π_i's data. Each method runs on 10 orders with 5 seeds, identical across methods.

P2, unseen group. One cell (S1) or condition (S6) is held out for testing; models are trained on the others.

P3, first-life pretraining. All datasets are mapped onto shared inputs (SOH history, calendar and cycle time, planned time to the target, operating conditions and chemistry; undefined conditions flagged as missing). Models are pretrained on the pool, which also gives the input scaling and the first EWC anchor; the seven S6 conditions then follow as a stream.

E. Models

Preprocessing (standardisation, median imputation with missing indicators, target scaling) is fitted on the first task and then frozen. The *MLP* has hidden layers of 128 and 64 ReLU units (dropout 0.1). The proposed backbone adapts the Temporal Fusion Transformer (TFT) [24]: a window of the $W{=}4$ most recent origins of the same cell and horizon (inputs only, never labels) passes through per-variable embeddings, variable selection by gated residual networks, an LSTM encoder [25] and interpretable two-head attention masked to past steps (Fig. 2; $d{=}8$, 12,708 parameters with the shared inputs).

Strategies. Fine-tuning trains on the current task only (lower bound); experience replay [26] stores B samples per finished task; joint training fits all tasks at once (upper bound); and Elastic Weight Consolidation (EWC) [27] adds, for each finished task j,

$$\frac{\lambda}{2}\sum_{p} F_{j,p}\left(\theta_p - \theta^*_{j,p}\right)^2, \tag{4}$$

with θ^*_j the parameters after task j and F_j the empirical diagonal Fisher information over up to 500 training samples. *TFT + EWC* is the proposed continual learner; trained on the residual (5) below, with shrinkage and intervals, it forms the proposed *Residual TFT-EWC* (Fig. 2). Training uses Adam [28], batches of 64, at most 200 epochs and early stopping on the current task's validation RMSE.

Reference forecasters are persistence ($\hat{\Delta} = 0$), linear extrapolation $\hat{\Delta}_{b,k,h} = r_{b,k}(t_{b,k+h} - t_{b,k})$, ridge regression (on

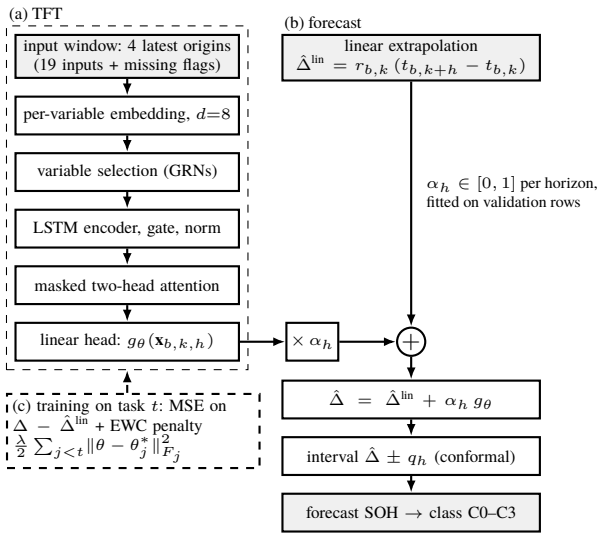


Fig. 2. Proposed model, Residual TFT-EWC. (a) The TFT maps a window of the four latest origins of a cell to a correction g_θ. (b) The correction is shrunk per horizon and added to linear extrapolation (5); conformal intervals give the reuse class. (c) Each task is trained on the correction with the EWC penalty (4).

planned usage only, F0 or F1) and histogram gradient boosting on F1.

Learned residual. Because linear extrapolation proved hardest to beat, a pre-registered follow-up trains the continual learners on the correction to it. With $\hat{\Delta}^{\text{lin}}_{b,k,h}$ the extrapolation above (zero when $r_{b,k}$ is missing), the model g_θ learns $\Delta_{b,k,h} - \hat{\Delta}^{\text{lin}}_{b,k,h}$ and the forecast is

$$\hat{\Delta}_{b,k,h} = \hat{\Delta}^{\text{lin}}_{b,k,h} + g_\theta(\mathbf{x}_{b,k,h}). \tag{5}$$

The residual and the direct (target Δ) forms are compared on the S6 stream with the P3 shared inputs, from scratch, for horizons 1–3 (E1) and 1–6 (E2).

Shift-adaptive EWC and literature backbones. A second pre-registered experiment lets the EWC strength follow the shift between groups (shift-adaptive EWC, SA-EWC): when task t starts, the penalty (4) of each finished task j is multiplied by $d_{j,t} = \big((\|\mu_t-\mu_j\|^2 + \|\sigma_t-\sigma_j\|^2)/D\big)^{1/2}$, the per-dimension 2-Wasserstein distance between diagonal Gaussians fitted to the two tasks' preprocessed training inputs, with λ_0 tuned as λ. BiLSTM [11], TCN [8], [11] and Transformer-encoder continual learners (TFT input window, within 20% of its parameters) and single-fit GPR [3], [4], random forest and SVR are run in residual and direct form.

Shrinkage and forecast intervals. A third pre-registered experiment calibrates the residual model after training, leaving its weights unchanged. Per horizon h, the learned correction is shrunk towards extrapolation by $\alpha_h = \min\big(1, \max(0, \sum p\,y / \sum p^2)\big)$, with p the predicted and y the observed correction on calibration rows. A split-conformal interval [29] $\hat{\Delta}\pm q_h$ takes as q_h the $\lceil (n+1)(1-a)\rceil$-th smallest of the n absolute calibration residuals at horizon h. Intervals at level $1-a$ (90% and 80%) are scored by coverage, width and the interval score [30] $2q_h + \frac{2}{a}\max(|\Delta-\hat{\Delta}| - q_h, 0)$ (pp; lower is better). On the stream both are refitted after each task on its validation rows; extrapolation gets intervals the same way.

Unseen conditions. A fourth experiment repeats P2 on S6 for the shrunk residual TFT + EWC, streamed over the six training conditions in the ten evaluation orders with the P1 tuned values; shrinkage and intervals are fitted once on their pooled validation rows. Origins with a fade rate (H^+) and without (H^-, a cell's first two checks, where extrapolation is persistence) are reported separately. A fifth experiment calibrates intervals by leaving one condition out: each training condition is predicted by a model fitted on the other five, the absolute residuals at all its origins, pooled over the six, give q_h (per horizon, optionally per stratum), and the interval is centred on the forecast of the model fitted on all six. This is the cross-validation interval discussed in [31] with conditions as folds; unlike their CV+, it has no finite-sample guarantee, so coverage is measured on held-out conditions. The forecaster is gradient boosting, the most accurate in P2; being deterministic, it must replicate across both horizon sets instead of seed sets. In a sixth experiment the held-out condition arrives over time: as each target is measured, gradient boosting is refitted with it and the half-width moves by $c\,q_h(e-a)$, $e=1$ for a miss (quantile tracking [32]; c chosen on the training conditions).

F. Hyperparameters, metrics and statistics

Hyperparameters are chosen on validation data from tuning orders disjoint from the evaluation orders: the learning rate per backbone ($3\cdot10^{-4}$ to 10^{-1}), then replay B (50–400) and EWC λ (10^0–10^5), extending any grid whose best value lies at an edge. From P3 on, ten tuning orders and the one-standard-error rule (the smallest grid value within one standard error of the best) are used.

After each step i, every task's test set is evaluated, giving $R_{i,j} = \text{RMSE}(\mathcal{D}^{\text{te}}_{\pi_j} \mid \theta_i)$. Following [33], [34], we report the final average error $\frac{1}{T}\sum_j R_{T,j}$, backward transfer $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - R_{j,j})$ and forgetting $\frac{1}{T-1}\sum_{j<T}(R_{T,j} - \min_{j\le i<T} R_{i,j})$ (positive values mean forgetting), the skill $1 - \text{RMSE}/\text{RMSE}_{\text{persistence}}$, and the agreement of the reuse class of the forecast SOH with the measured class (C0 $>$ 95%, C1 90–95%, C2 80–90%, C3 $\le$ 80%, fixed in advance). The classes screen for capacity fade relative to the first valid check; they do not by themselves establish suitability for an application.

For P1 and P2 we also report further regression metrics, including R^2_Δ of the change since the origin (≤ 0 for persistence), and class metrics such as macro-F1 and the Matthews correlation coefficient (MCC); the full sets are in the repository (decision D30). Means carry 95% bootstrap intervals [35]. Methods are compared by two-sided Wilcoxon signed-rank tests [36] on matched runs or held-out groups, with rank-biserial effect sizes [37] and Holm correction [38]. A difference d is *replicated* only if it is significant in the main seed set (0–4) and in a replicate set (5–9), with the same sign and $|d| \ge 0.05$ pp in both; $|d| < 0.05$ pp in both means

practically equivalent. The threshold is about half the label shift caused by the pulse-charge definition.

G. Implementation and availability

The pipeline uses Python 3.11, NumPy 2.4, pandas 3.0, scikit-learn 1.9 and PyTorch 2.14.1 (P1 and P2 were first run on an Intel Xeon with an unrecorded PyTorch version and re-run on an Intel Core 5 210H), with one CPU thread and seeded, deterministic algorithms. Every result file records configuration, seeds, library versions and the Git commit, and unit tests cover input causality, split purging and label exclusion. The datasets are at doi:10.17605/OSF.IO/8JNR5 (S1), 10.57745/OLBXKT (S6), 10.35097/1969 (S7), 10.5281/zenodo.10963339 (S8) and 10.5281/zenodo.6405084 (S9); code, decision log, pre-registrations and results are at https://github.com/TarunSitaraman/residual-tft-ewc.

IV. Results

Summary. Neither traditional forecaster is reliable in both settings (Table III; pairs such as 0.32/0.31 give the main and replicate seed sets). Gradient boosting, which sees every condition's data at once, is best on an unseen condition, yet 0.13–0.41 pp less accurate than the proposed model on the continual stream (replicated; post hoc for the shrunk model). Linear extrapolation is the mirror image: level with the proposed model on the stream, 0.43–0.68 pp less accurate on an unseen condition (replicated). The proposed model is level with or better than each where the other fails. Held to the same constraint (one condition at a time, no earlier data kept), gradient boosting refitted on the latest condition is slightly more accurate for horizons 1–3 (0.06/0.08 pp, replicated) and not distinguishable for 1–6; warm-started boosting cannot use the usage inputs after the first condition and is far worse (supplement).

Data. S1 diagnostics track the current SOH (median within-cell $\rho = 0.97$ for the main incremental-capacity peak) but not the change to the next RPT (median $|\rho| \leq 0.3$); every S1 sample stays in class C0, while all S6 cells cross from C0 to C3.

A. Continual stream (P1)

S1. The methods are practically equivalent: TFT + EWC reaches 0.462 pp (skill 0.27), level with gradient boosting, TFT fine-tuning and MLP replay (0.466–0.475 pp; Table SIV of the supplement), ridge on the planned usage alone reaches 0.450 pp, and forgetting is small throughout (0.016–0.040 pp).

S6. When tasks differ in usage, forgetting is substantial (0.18–0.46 pp). EWC lowers the TFT's error relative to fine-tuning (0.677 against 0.883 pp), though which difference is significant changed between runs (§IV-B); the MLP beats the TFT in both runs (0.531–0.540 pp), joint training puts the TFT's continual-learning cost at about 0.25 pp, and linear extrapolation (0.343 pp) and gradient boosting (0.372 pp) are best.

B. Unseen cell or usage condition (P2)

Gradient boosting is best on both datasets (Table IV). On S6 the 100% depth-of-discharge condition is the hardest for every method (1.8–2.4 pp). With six or seven held-out groups, no difference with the TFT survives Holm correction.

In the further metrics, gradient boosting has the largest R^2_Δ (0.54 on S1, 0.74 on S6), while extrapolation forecasts the change no better than its mean (-1.31, -0.05). The S6 reuse class gives the same order (Table V; MCC 0.93 for gradient boosting, 0.72 for persistence).

Reproducibility. A re-run of P1 and P2 in a pinned environment, on data rebuilt from the raw archives, reproduced every baseline, every P2 result and the ranking of methods. The neural models moved by at most 0.059 pp, enough to change the significance of differences of 0.005–0.014 pp: neural training is deterministic per platform, not across platforms. This motivated the stricter tuning and the replication rule used from P3 on.

C. First-life pretraining (P3)

Pretraining on the first-life pool (36,711 samples, against 2,409 in S6) does not help. Both backbones are better from scratch on average (TFT 0.401/0.415 pp against 0.509/0.591 pp pretrained with EWC, significant only in the replicate set), and linear extrapolation (0.343 pp) remains the most accurate method, on S6 and on the pool's own test split (supplement).

Why the TFT improved. A post hoc ablation, repeated on the replicate seeds, explains why the TFT from scratch improves from 0.891 to 0.401 pp with the shared inputs: dropping the 17 diagnostic features (0.17–0.35 pp) and adding the calendar time between checks (0.19–0.31 pp), both replicated ($p_{\text{Holm}} \leq 0.002$).

D. Learned residual on linear extrapolation

Learning the residual (5) (Fig. 3; Table SIII of the supplement) did *not* meet the pre-registered criterion of beating linear extrapolation: the two are practically equivalent for horizons 1–3, and for 1–6 extrapolation is slightly better on average (not replicated); per horizon it is better at $h = 1$–2 (replicated), and the gap closes by $h = 5$–6. It does make the TFT competitive: 0.17 pp (E1) and 0.24–0.30 pp (E2) more accurate than the direct form and ahead of gradient boosting, all replicated. For horizons 1–6, where it forgets more, EWC lowers its forgetting from 0.28/0.22 to 0.09 pp and its backward transfer from 0.18/0.14 to 0.03 pp, replicated; its effect on the final error is not. For horizons 1–3, EWC and fine-tuning are practically equivalent. *Ageing-cycle inputs.* The 21 Spark-extracted inputs (Section III-B), added as pre-specified, make the residual TFT + EWC worse: 0.480/0.501 against 0.339/0.314 pp, and worse than linear extrapolation, both replicated.

E. Shift-adaptive EWC and literature baselines

The second pre-registered criterion, for SA-EWC (Table SIII of the supplement), was *not* met: although the shift weights varied (0.05–1.14), SA-EWC and fixed λ are practically equivalent in final error and forgetting (differences of at most

TABLE III

THE PROPOSED MODEL AGAINST THE TWO STRONGEST TRADITIONAL FORECASTERS ON S6, IN BOTH SETTINGS. TOP: TEST RMSE (PP); PROPOSED MODEL: MEAN OVER 50 RUNS, MAIN/REPLICATE SEED SETS; OTHERS: SINGLE DETERMINISTIC FITS (BOLD: BEST). BOTTOM: PROPOSED MINUS BASELINE (PP, MAIN/REPLICATE; NEGATIVE FAVOURS THE PROPOSED MODEL) AND THE VERDICT OF THE REPLICATION RULE (BETTER/WORSE: SIGNIFICANT AFTER HOLM IN BOTH SEED SETS, SAME SIGN, AT LEAST 0.05 PP; EQUIV.: BELOW 0.05 PP IN BOTH; N.R.: NOT REPLICATED). *POST HOC FOR THE SHRUNK MODEL; REGISTERED FOR THE UNSHRUNK MODEL (D23), WITH THE SAME VERDICT. †UNDER THE CONTINUAL CONSTRAINT OF THE PROPOSED MODEL (CONDITIONS ONE AT A TIME, EARLIER DATA NOT KEPT); P2 ONLY (D32).

	Continual stream (P1)		Unseen usage condition (P2)	
	h=1–3	h=1–6	h=1–3	h=1–6
Residual TFT-EWC (proposed)	**0.32/0.31**	0.50/0.49	0.92/0.94	1.62/1.65
Linear extrapolation	0.34	**0.48**	1.37	2.30
Gradient boosting, all conditions stored	0.45	0.90	**0.54**	**1.05**
Gradient boosting, latest condition only†	–	–	0.86	1.59
Persistence	1.64	2.67	2.82	4.90
Proposed minus baseline, verdict				
vs. linear extrapolation	−0.02/−0.04 equiv.	+0.01/+0.01 equiv.	−0.45/−0.43 better	−0.68/−0.65 better
vs. gradient boosting, all stored	−0.13/−0.14 better*	−0.40/−0.41 better*	+0.39/+0.41 worse	+0.57/+0.60 worse
vs. gradient boosting, latest only†	–	–	+0.06/+0.08 worse	+0.03/+0.06 n.r.

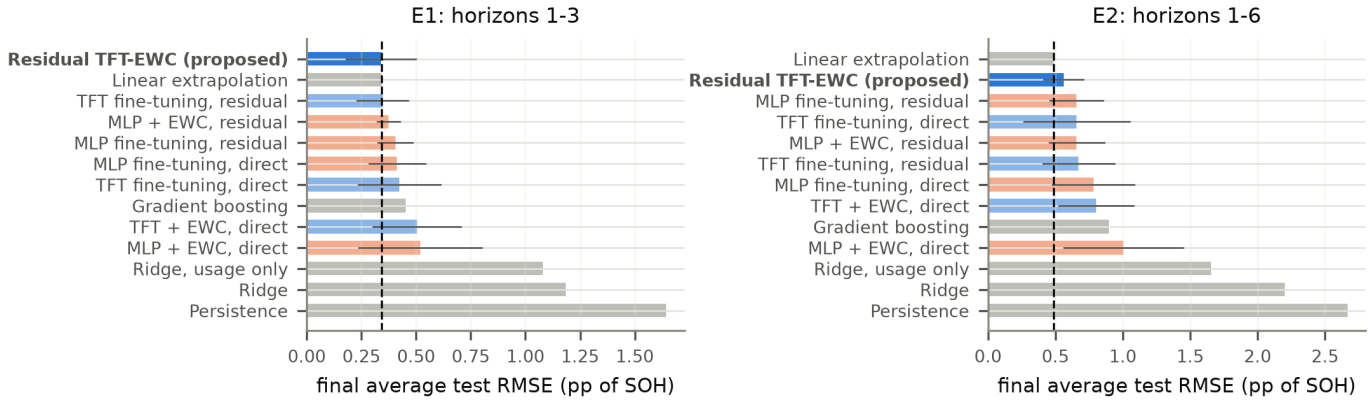


Fig. 3. S6 continual stream: final average test RMSE (pp), mean and s.d. over the 50 runs of the main seed set; bold, blue: the proposed residual TFT-EWC; dashed: linear extrapolation. Replicate set, forgetting and literature backbones: Table SIII of the supplement.

TABLE IV

PROTOCOL P2 (UNSEEN BATTERY IN S1, UNSEEN USAGE CONDITION IN S6): MEAN TEST RMSE (PP) OVER HELD-OUT GROUPS, RANGE OVER GROUPS. REUSE-CLASS METRICS OF THE SAME FORECASTS: TABLE V.

Data	Method	RMSE	Range
S1	Persistence	0.545	0.436–0.688
S1	Linear extrapolation	0.511	0.284–0.725
S1	Ridge, usage only	0.301	0.172–0.409
S1	Ridge, F0	0.418	0.158–0.779
S1	Ridge, F1	0.320	0.161–0.477
S1	Gradient boosting, F1	0.219	0.091–0.299
S1	MLP	0.285	0.149–0.389
S1	TFT	0.292	0.199–0.377
S6	Persistence	2.817	2.025–4.327
S6	Linear extrapolation	1.372	0.856–1.669
S6	Ridge, usage only	1.177	0.870–2.288
S6	Ridge, F0	0.859	0.598–1.836
S6	Ridge, F1	1.137	0.588–2.188
S6	Gradient boosting, F1	0.710	0.395–1.839
S6	MLP	0.821	0.476–2.225
S6	TFT	0.868	0.551–2.390

TABLE V

S6, P2: REUSE CLASS (C0–C3) AND END OF FIRST LIFE (EOL, SOH $\leq$ 80) OF THE FORECAST SOH, HELD-OUT SAMPLES POOLED, MEAN OVER SEEDS; F1 MACRO, κ_w LINEAR-WEIGHTED. ALL METRICS AND CLASS COUNTS: TABLE SII OF THE SUPPLEMENT.

Method	Acc.	F1	MCC	κ_w	EoL F1	EoL AUC
Persistence	0.858	0.661	0.723	0.809	0.962	0.998
Linear extrapolation	0.939	0.777	0.876	0.909	0.989	0.999
Ridge, usage only	0.954	0.851	0.905	0.932	0.989	0.999
Ridge, F0	0.963	0.881	0.922	0.942	0.990	0.999
Ridge, F1	0.958	0.852	0.912	0.935	0.986	0.999
Gradient boosting, F1	0.966	0.893	0.928	0.948	0.989	0.999
MLP	0.962	0.871	0.921	0.942	0.989	0.999
TFT	0.963	0.882	0.923	0.943	0.989	0.999

0.03 pp). For horizons 1–3, BiLSTM + EWC (direct, 0.323 pp, and residual) and the residual Transformer + EWC are practically equivalent to the residual TFT; for 1–6 it beats the residual BiLSTM and TCN (replicated; not the Transformer). Every single-fit regressor is worse than the residual TFT + SA-EWC (replicated), except the residual SVR for horizons 1–6.

F. Shrinkage and intervals on the stream

Shrinking the correction per horizon makes the residual TFT + EWC practically equivalent to extrapolation at every horizon (0.318/0.308 against 0.343 pp for horizons 1–3,

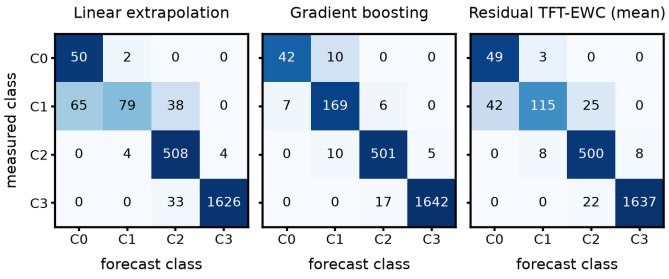


Fig. 4. Reuse-class confusion on unseen S6 conditions (Table SVI of the supplement): sample counts (proposed model: mean over 50 runs); shading: share of the measured class.

0.498/0.493 against 0.485 pp for 1–6), removing its loss at $h = 1$–2, but the criterion was again not met. Its 90% intervals are narrower than extrapolation's but cover too little (0.71–0.81, against 0.92–0.95), so the interval scores are equal. Pretraining the residual model on the first-life pool makes it worse for horizons 1–3 (by 0.22 pp, replicated) and does not help for 1–6.

G. Residual model on unseen conditions

On unseen conditions (Table III; Table SV of the supplement), the shrunk residual TFT + EWC beats extrapolation, as registered, at origins with a fade rate (by 0.12 pp for horizons 1–3 and 0.18–0.20 pp for 1–6, replicated), its first win over extrapolation. At a cell's first two checks, extrapolation reduces to persistence and loses by over 2 pp. Gradient boosting, however, is 0.39–0.41 pp (horizons 1–3) and 0.57–0.60 pp (1–6) more accurate than the TFT (replicated), on six of the seven conditions; only on the 100% depth-of-discharge condition is the TFT better. The same residual TFT trained jointly improves on its continual form by 0.32–0.44 pp (replicated), and shrinkage makes no practical difference.

The capacity-based classes follow the same ranking (Table SVI of the supplement, Fig. 4): the proposed model assigns the measured class to 95.5% of held-out samples (94.4–96.8% over the 50 runs), against 93.9% for extrapolation and 85.8% for persistence, while ridge (96.5%) and gradient boosting (97.7%) do better. All errors are between neighbouring classes; the most common is the risky one for reuse, a measured C1 sample forecast as C0 (65 of 182 for extrapolation, 42 for the proposed model, 7 for gradient boosting).

Intervals on unseen conditions. Intervals calibrated on validation rows of the training conditions cover too little on the held-out one (0.66–0.68 for the TFT and 0.74–0.76 for extrapolation, at 90%). The calibration rows are much of the cause: calibrating extrapolation, which has no fitted parameters, on every origin of those conditions raises its coverage to 0.89–0.90. Gradient-boosting intervals calibrated by leaving one condition out reach the registered 0.85 (0.90) with the lowest interval scores (3.2 and 6.7 pp), though their advantage over calibrated extrapolation is significant only for horizons 1–3 (not established). Pooled coverage hides one failure: the 100% depth-of-discharge condition, outside the others' range, is covered only 0.48 and 0.38 (Fig. S8, supplement).

Refit and tracking. As the held-out condition's checks arrive, refitting gradient boosting and tracking the half-width lower its mean 90% interval score by 29%, from 3.22 to 2.29 pp (h=1–3; six of seven conditions, $p = 0.047$) and from 6.67 to 4.74 pp (h=1–6; seven of seven, $p = 0.016$), with pooled coverage 0.90–0.91. On the 100% depth-of-discharge condition coverage rises from 0.48/0.38 to 0.72/0.58, short of the registered 0.80. Then 87–93% of held-out intervals lie within one reuse class, 99.2–99.7% of them correctly.

V. Discussion

Why the proposed model does not win. On the stream, TFT + EWC never beat linear extrapolation: fade is smooth, the groups are few and small (21–27 and 135–225 training samples per task in S1 and S6), and the TFT's attention over its history is nearly uniform. EWC helps where there is something to protect (S6, not S1), most clearly on the residual target, where it cuts forgetting over six horizons by about two thirds; a shift-adaptive penalty added nothing, and at horizons 1–3 a BiLSTM or Transformer with EWC is as accurate. On unseen conditions the continual constraint itself is the main cost (joint training is 0.32–0.44 pp more accurate). With data this small, storing every earlier group is cheap, so the constraint that motivates continual learning did not bind.

What matters. Building on the best simple forecaster mattered most; the ablation points to inputs, not architecture: diagnostic features hurt the TFT, and the calendar time between checks helped, consistent with fade per cycle rising with that time at 50 °C in all 20 S6 cells. For intervals, the calibration rows mattered as much as the model: only residuals from held-out conditions reproduce the error on a new one.

Reproducibility and reuse. As significance changed across platforms (Section IV-B), we recommend recording the environment and requiring replicated differences above a practical threshold. The reuse classes are a screening aid, not a suitability verdict: they rest on relative capacity alone, whereas reuse also depends on power capability, safety, absolute capacity and cost. They are informative only when the horizon crosses class boundaries, and for a population unlike any seen, such as the 100% depth-of-discharge condition, neither forecasts nor intervals can be trusted, which pooled coverage hides.

Limitations. The task was changed to forecasting after the first dataset was inspected; the second-life datasets are small; S6 is aged at 50 °C and its "second life" continues the same protocol; SOH is relative to each cell's first valid measurement; forecasts are conditional on planned usage; and the ablation and every later experiment were decided after the main results (all but the ablation committed before being run). The single-fit regressors see all tasks at once, and the direct literature backbones were run for horizons 1–3 only. The TFT hyperparameters for unseen conditions were tuned with validation rows of all seven S6 conditions, all from one laboratory. Leave-one-condition-out intervals were run for gradient boosting and extrapolation only.

VI. Conclusion

We evaluated a TFT with EWC as a continual learner for forecasting the SOH of second-life cells. Learning a shrunk correction to linear extrapolation made it as accurate as extrapolation and more accurate than gradient boosting on the stream, and more accurate than extrapolation on unseen usage conditions, where gradient boosting with all conditions stored and the same TFT trained jointly were clearly better: at this data scale, learning continually cost more than it gave. Conformal intervals held on unseen conditions only when calibrated on conditions held out from fitting, and not outside the training range; refitting and tracking cut the interval score by 29%. Future work should test continual learning where earlier data cannot be stored and populations are larger, calibrate intervals across laboratories and chemistries, and add power and safety criteria to the screening.

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